

Neighborhood Choice

EC3335, Set 06

Chen Xu

Spring 2027

"Love thy neighbor as yourself, but choose your neighborhood."

-Louise Beal

Housekeeping

Assignments:

- **PS01** was due last night
- **Reading**
 - finish up to chapter 5 before the midterm

Midterm: Wed, Nov 2

Introduction to neighborhood choice

We have a fairly simple model of **residential choice** (rental prices)

Q: What factor(s) in the model determine housing demand?

A: Bid-Rent model assumes commuting costs are the **only factor**

- this **unreasonable** assumption isolates particular economic factors

What factors influence neighborhood decision choices?

Examples:

- Schools
- Demographics
- Tax rates
- Public safety
- Air quality
- Natural beauty

Neighborhood choice: Amenities

Definition: **Amenity**

An *amenity* is a **location-specific** consumption good

- Beaches
- Weather
- Public transport
- Parks
- Restaurants
- Recreation

Different types of **amenities**

- Some are nonrival[†]: Theaters, public transport
- Some are nonexcludable^{††}: Parks
- Some are both nonrival and nonexcludable: National defense, sports teams, fireworks

[†] Nonrival goods: Accessible by all; usage does limit subsequent use

^{††} Nonexcludable goods: Impossible to exclude other from consuming

Neighborhood choice: Amenities

Two more refined definitions:

(i) Exogenous Amenities: Location-specific consumption good that exist **are not** influenced by where people decide to live

- **Exogenous:** "Determined outside of the model" (fall from the sky)
- Weather, geographic characteristics

(ii) Endogenous Amenities: Location-specific consumption goods that **are** influenced by location decisions of individuals

- **Endogenous:** "Determined within the model"
- School quality, crime, pollution

Neighborhood choice: Amenities

To determine whether or not an amenity is **exogenous** (**endogenous**):

"Will choosing to live here impact the amenity?"

- **Exogenous** *Beaches exist regardless whether people live near by*
- **Endogenous** *Crime is a function of the individuals in the area*

Questions regarding differences between **EXOGENOUS** and **ENDOGENOUS?**

Neighborhood choice

Previously we explored (broadly) a cities shape

- modeled where different sectors of the economy choose to locate

But we made an assumption that everyone was the **same**

*Why are city neighborhoods so **heterogeneous**?*

*What economics factor influence **neighborhood sorting** within a city?*

Model this **heterogeneity** by considering a public good: **Public parks**

Neighborhood choice + Sorting for public goods

Demand for public goods

Consider a simple sorting model for a single, non-rival public good

Model a three-person city with one public good: **Public park**

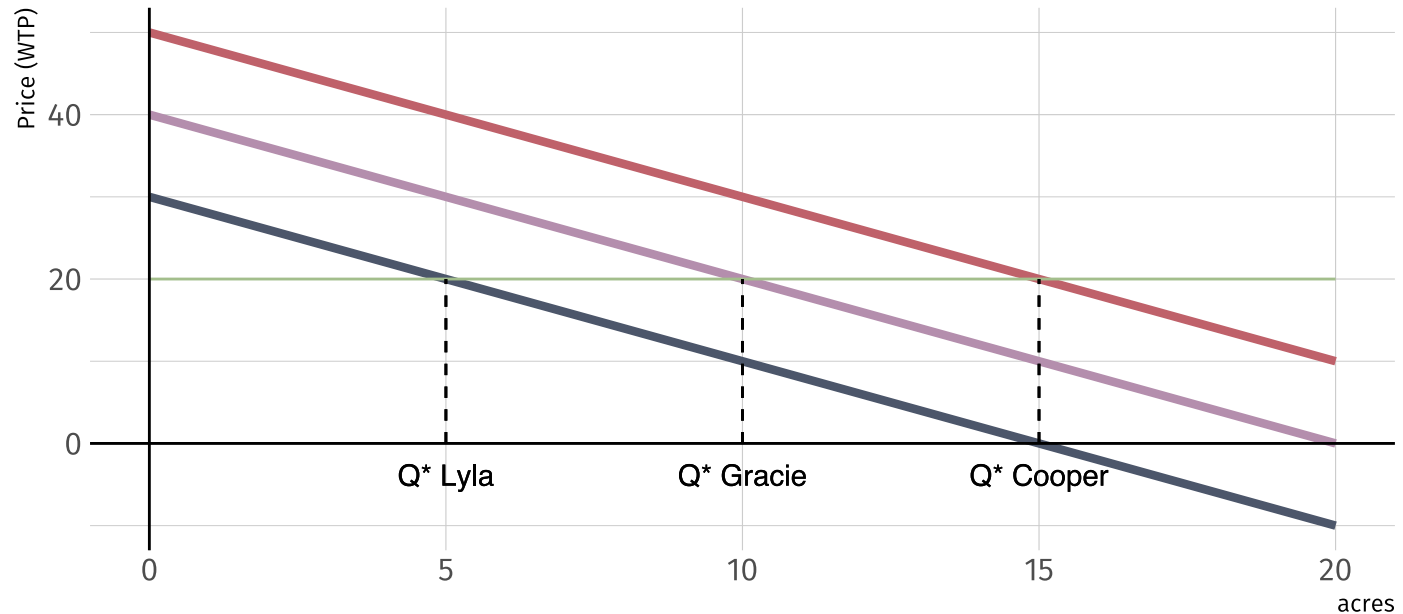
- Cost **\$60 per acre** to build
- Cost is shared equally across all three citizens: **\$20 per acre** each

Of the three citizen, demand for the park varies:

- Low demand: **Lyla** : $P = 30 - 2 * \text{acres}$
- Mid demand: **Gracie** : $P = 40 - 2 * \text{acres}$
- High demand: **Cooper** : $P = 50 - 2 * \text{acres}$

Together they must **vote** for one park size in a binary election

Demand for public goods



If a first past the post election system is used, who wins their preferred park size? Who loses?

Demand for public goods: Majority rule

Under **majority rule** Gracie's optimal park wins

Why?

Election	10 acre votes	Other votes
5 acres vs 10 acres	Gracie and Cooper	Lyla
15 acres vs 10 acres	Gracie and Lyla	Cooper

Gracie is the **median voter**

- Splits the voting public in half

Alternative to majority rule

Majority rule always leave two citizens unhappy: **Cooper** and **Lyla**

Suppose an *alternative* is to form new municipalities

Consider a new metropolitan area with **3** municipalities with 3 citizen

- Each district votes on their own park
- Each citizen knows each other's preferences

Key assumption: Citizens pick which municipality to live in

- Lylaville: 5 acre park
- Gracity: 10 acre park
- Cooperstown: 15 acre park

Alternative to majority rule

By **voting with their feet** each citizen sorts themselves into homogenous communities with their preferred public good allocation (park size)

Now our city has three neighborhoods with homogenous types

- Accommodates diversity in demand

Is reality this simple?

Nope

Let's add another layer of complexity: Taxes

Alternative: Property tax

Up to this point, funding for the park is financed with a **head tax**

More realistic to model neighborhood sorting using **property taxes**

- Allow for variation in preferences + property values:
 - The higher your property value, the more taxes you pay for the park
 - $\tau = PV * 10$
 - **3** different property values: 2, 10, 24
 - **9** combos: Low-Low, Low-Mid, Low-High, Mid-Low...High-High

Outcome	Tax rate per dollar in PV	Small PV	Mid PV	Big PV
Mixed municipality	\$10	\$20	\$100	\$240
Exclusive small PV	\$60	\$120	-	-
Exclusive mid PV	\$12	-	\$120	-
Exclusive big PV	\$5	-	-	\$120

Alternative: Property tax

Outcome	Tax rate per dollar in PV	Small PV	Mid PV	Big PV
Mixed municipality	\$10	\$20	\$100	\$240
Exclusive small PV	\$60	\$120	-	-
Exclusive mid PV	\$12	-	\$120	-
Exclusive big PV	\$5	-	-	\$120

Citizen of PV similar type have incentive to sort together to reduce tax

In equilibrium citizen will form **9 different neighborhoods**

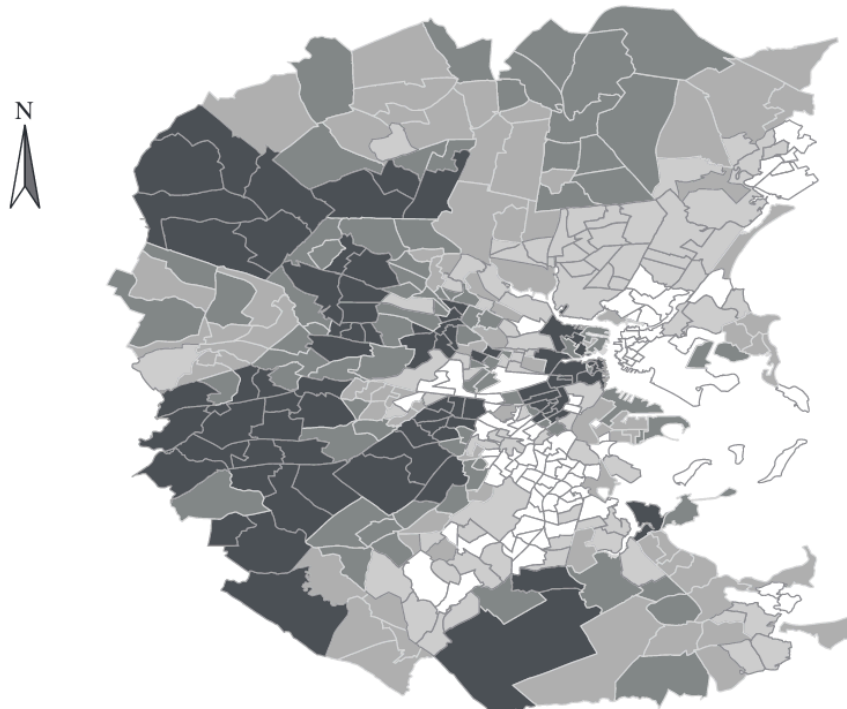
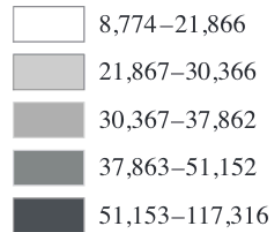
Generates a fragmented system of local government in a metro area

- Negative implications arise from this sorting

Neighborhood sorting: Income

MAP 8-1 Income Segregation: Boston

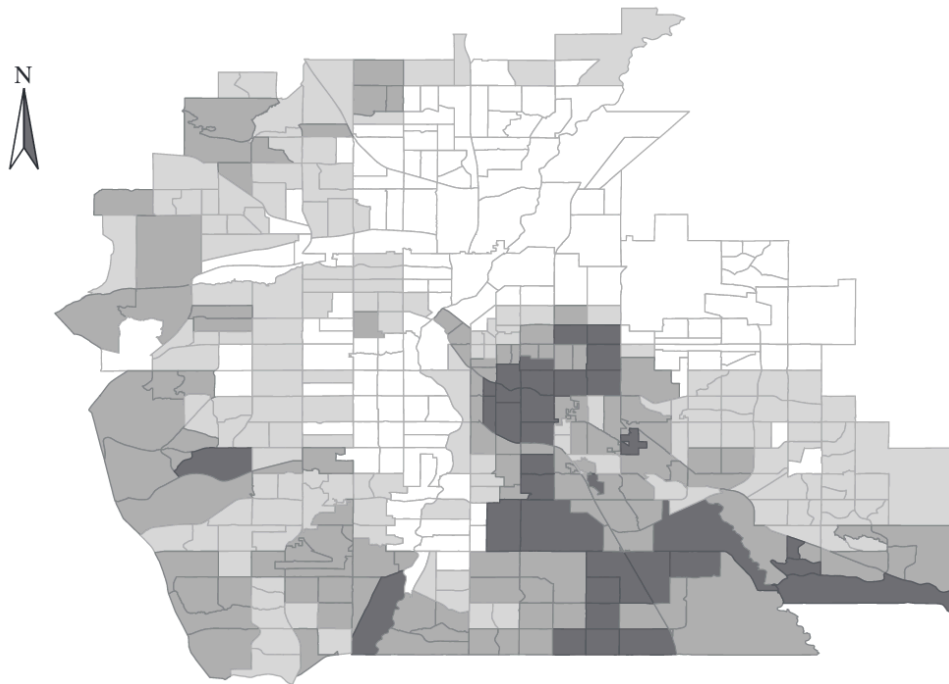
Per-Capita Income (\$)



Neighborhood sorting: Education

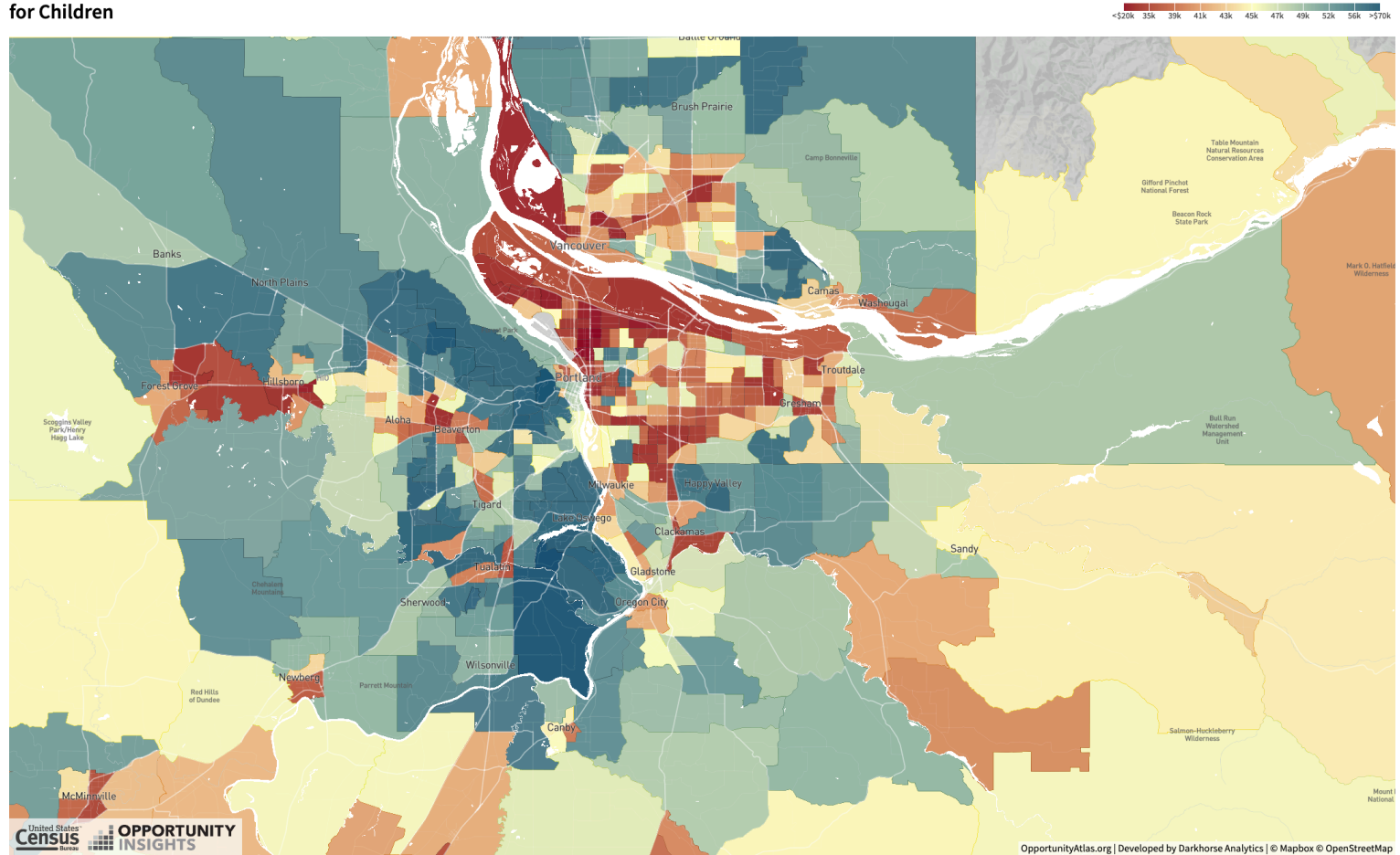
MAP 8-2 Segregation with Respect to Educational Attainment: Denver

College Degrees per 1,000



Neighborhood sorting: Atlas

Household Income
for Children



Neighborhood sorting: Externalities

Neighborhood sorting: Externalities

Do you **fully internalize** the costs and benefits of where you decide to live?

- *Is your choice of neighborhood free from externalities?*

Nope.

Examples?

- Social networks
- Jobs
- Good schools
- Culture
- Noise
- Drug use
- Litter
- Pollution

Neighborhood externalities tend to be massively important for **youth**

- Peer effects
- Role models

Neighborhood sorting: Externalities

When your neighbor...

- remodels their kitchen...
- gives the scoop on a new job...
- tells you how to apply for a fraudulent PPP loan...
- does a bunch of drugs...
- makes a ton of noise...
- is a bad role model to your children...

... *do you pay them?*

... *do they pay you?*

Positive externalities general increase with income and education level

- more desirable; higher demand

What does this *imply?*

Becker-Murphy model

Becker-Murphy model

In real life households compete for places in **desirable** neighborhoods

In the Becker-Murphy model we consider this competition

- *land always goes to the highest bidder*

Focus on **positive** externalities for now

- assume these increase with income and education
- ie desirability is a function of number of high income neighbors

Consider a model where:

- Two neighborhoods, A and B (80 lots each)
- Infinite number of households on the market
- Only difference between the neighborhoods is income mix

Becker-Murphy model

Individual choices to move are determined by the *rent premium*

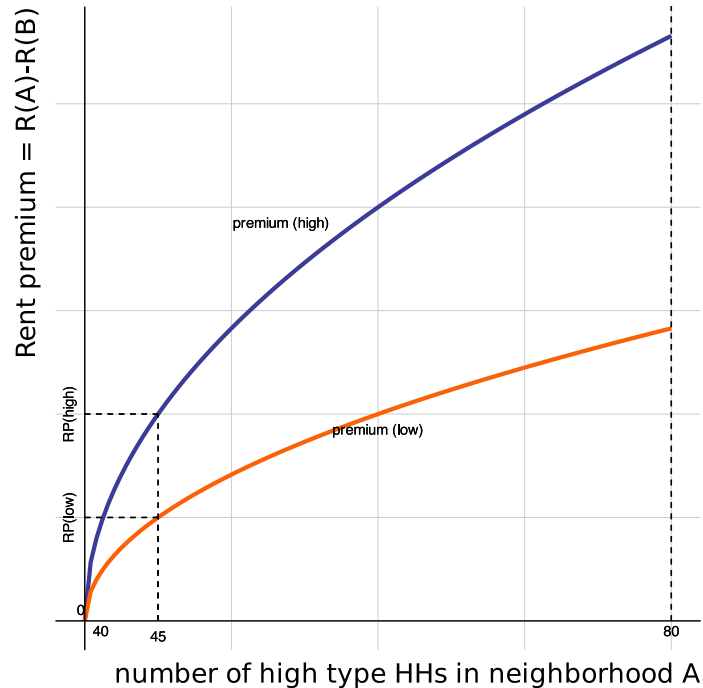
Rent Premium: Difference in rent between A and B

- $RP = R(A) - R(B)$ (for neighborhood A)

In this model:

- Consider two types of workers: **HT**, **LT**
- **RP** for workers (may) differ by type: $RP_h \neq RP_l$
 - ie benefit of living close to high types might vary by type
- land will be allocated to the highest bidder
- rental prices are **homogeneous** within a neighborhood

Integrated vs Segregated Eq



If 5 high types move into A

- $RP(high) > RP(low)$
- favorable mix of neighbors
- HT WTP for A increases
- cascading effect
- Neighborhood A is only high income HH's **segregated eq**
 - Slope **blue line** > slope **red line**

Recall **A02**: Self-reinforcing changes lead to extreme outcomes

Eq Defn

Recall: an **locational eq** is a point at which no one wants to move

- in this model, it occurs when the rent premiums are equal

This always holds when the rent premium curves intersect

May also occur when they do not (full segregation)

- If the RP for the group listed on the axis is **higher** then this will also be an equilibrium because **there is no tendency for change**
- If the RP for the group listed on the axis is **lower** then population dynamics move away from this point

Stable vs Unstable Eq

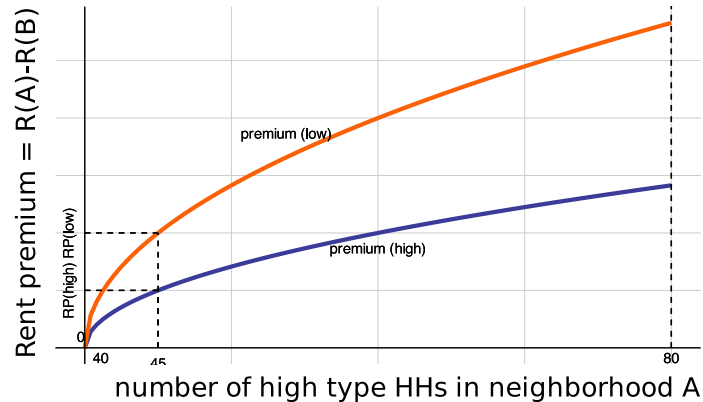
An equilibrium is **stable** if a small movement away will encounter self - **correcting** forces

- An equilibrium is stable if when you move away from it, the pop. dynamics push you back to where you came from
- Physical ex: Funnel

A equilibrium is **unstable** if a small movement away will encounter self - **reinforcing** forces

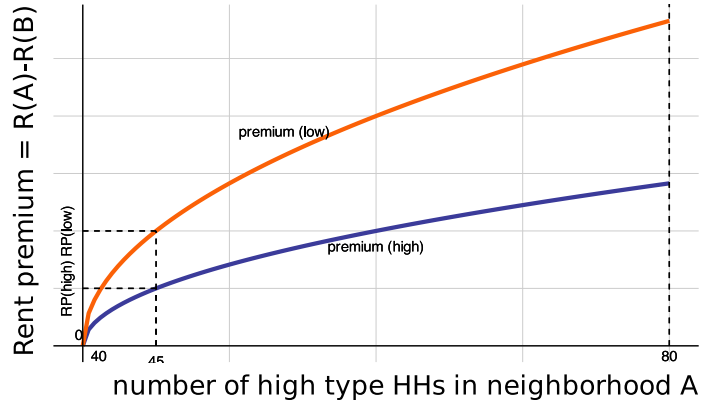
- That is, an equilibrium is unstable if when you move away from it, the population dynamics push you even farther than where you came from
- Physical ex: Stacking golf balls

Integration Eq



- Is the story the same here?
- Now, a small movement of high income HH's into A means $RP(\text{High}) < RP(\text{low})$
- So we get pushed back to the initial equilibrium. In this case, integration is the **only equilibrium**
- Furthermore, integration is a **stable equilibrium**

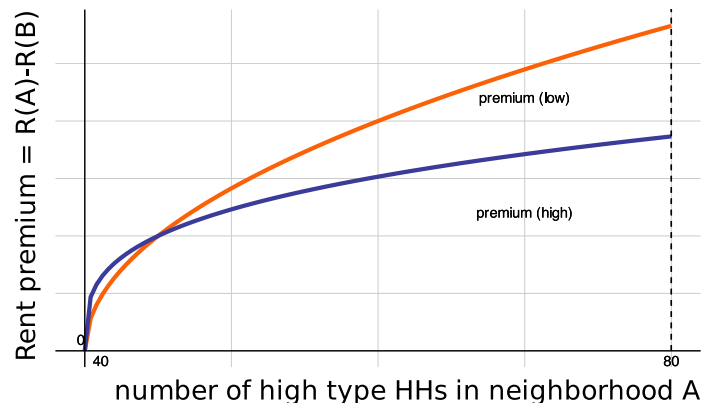
Integration Eq



Note: 80 high income HH's in A is not an EQ because $RP(\text{low}) > RP(\text{high})$. So low incomes will outbid highs and move in

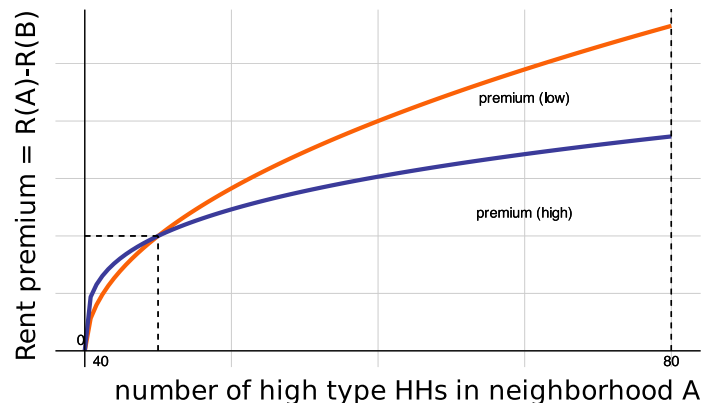
- Is the story the same here?
- Now, a small movement of high income HH's into A means $RP(\text{High}) < RP(\text{low})$
- So we get pushed back to the initial equilibrium. In this case, integration is the **only equilibrium**
- Furthermore, integration is a **stable equilibrium**

Mixed Eq



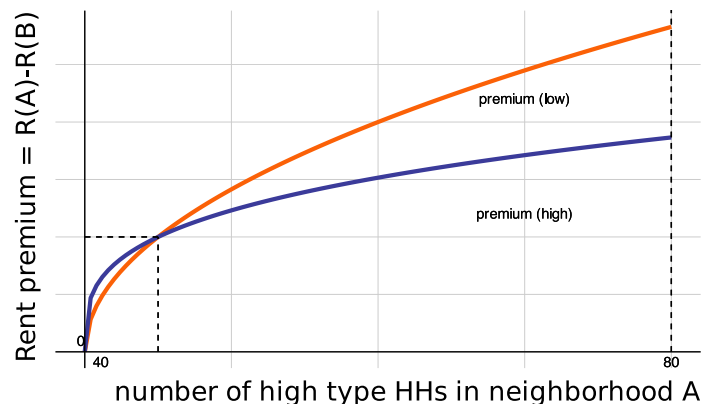
- What about a story like this?
- Integration eq (40 of each type in each nbhd) is still an equilibrium. Is it **stable**?
- No. A small deviation away means $RP(high) > RP(low)$. So highs outbid lows until $RP(high) = RP(low)$ at 45 highs in A and 35 lows.
- Is 45 highs in A stable?

Mixed Eq



- What about a story like this?
- Integration eq (40 of each type in each nbhd) is still an equilibrium. Is it **stable**?
- No. A small deviation away means $RP(high) > RP(low)$. So highs outbid lows until $RP(high) = RP(low)$ at 45 highs in A and 35 lows.
- Is 45 highs in A stable? Yes (you think about why)

Mixed Eq



- **Note:** Full segregation here is *not* an equilibrium for a similar reason to the last example

- What about a story like this?
- Integration eq (40 of each type in each nbhd) is still an equilibrium. Is it **stable**?
- No. A small deviation away means $RP(high) > RP(low)$. So highs outbid lows until $RP(high) = RP(low)$ at 45 highs in A and 35 lows.
- Is 45 highs in A stable? Yes (you think about why)

A Heuristic

- 1) Draw a vertical dashed line at every intersection point
- 2) For every region between the vertical dashed lines, it must be the case that one of the rent premium curves is above the other
 - If the rent prem curve for the group listed on the axis is **higher**, then this group will increase in number. Draw rightward arrows on the axis
 - If the rent prem curve for the group listed on the axis is **lower**, then this group will decrease in number. Draw leftward arrows

A Heuristic

3) If there are rightward arrows pushing toward 100% in one nbhd, then 100% (complete segregation) is an eq even if the rent prem curves do not intersect there

4) For every eq. value, look at its immediate vicinity

- If there are arrows moving towards it, it is a **stable eq**
- If there are arrows moving away from it on one or both sides, it is a **unstable eq**

Neighborhood Sorting

Externalities for kids:

- Good/bad role models as adults
- Classmates in school: focused vs disruptive

Externalities for adults:

- Positive: job information, property valuation
- Negative: property values

In general: positive externalities increase with income and education level.
Why?

Neighborhood Sorting

These externalities give rise to the following questions:

- (i)** Who gets desirable neighbors?
- (ii)** Will there be segregated or integrated neighborhoods?
- (iii)** Will there be sorting or mixing with respect to income, age, race, or some combination of those factors?
- (iv)** What are the implications for the price of land in various neighborhoods?

End of MT material